QUANTITATIVE FINANCE · RESEARCH & ANALYTICS

Monte Carlo Portfolio Risk — Analysis Report

Simulation results, interpretation, risk metrics, and business value

1. Executive summary

This project uses Monte Carlo simulation to explore a range of possible one-year outcomes for a hypothetical equal-weighted portfolio of five US technology stocks: AAPL, AMZN, GOOGL, MSFT, NVDA. It generates 10,000 simulated paths, each spanning 252 trading days, starting from $10,000.00.

The model estimates a mean terminal value of $13,972.82 and a median terminal value of $13,440.80. The modeled probability of ending below the initial investment is 14.45%. The 95% VaR loss estimate is $1,452.91, while the 95% Expected Shortfall estimate is $2,323.95.

These are the outputs of the configured model, not predictions or guaranteed investment outcomes. The numerical values depend on the sample period, return distribution assumptions, portfolio weights, and random seed.

2. What was analysed

Historical input data

- Historical start date: 2016-10-17
- Historical end date: 2026-10-16
- Daily return observations: 2,513
- Assets: AAPL, AMZN, GOOGL, MSFT, NVDA
- Starting portfolio value: $10,000.00
- Portfolio weights: equal weights across five stocks

- Rebalancing: daily
- Simulation seed: 42

The price series is converted to daily log returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right)$$

Symbol	Name	Description
$r_{i,t}$	Log return	Daily log return of asset i on day t
$P_{i,t}$	Asset price	Price of asset i at time t
$P_{i,t-1}$	Asset price (previous)	Price of asset i at time $t-1$
$\ln$	Natural logarithm	Logarithm to base e

For each asset, the sample mean and the joint covariance matrix are estimated from historical daily log returns. The covariance matrix captures both each asset's variability and the estimated co-movement between assets.

Correlated simulation

For each simulated day, the model draws a vector of asset log returns from a multivariate normal distribution:

$$\mathbf{r}_t \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Symbol	Name	Description
$\mathbf{r}_t$	Return vector	Simulated daily log returns of all five assets on day t
$\mathcal{N}$	Multivariate normal	Normal distribution from which daily log returns are drawn
$\boldsymbol{\mu}$	Mean vector	Historical sample mean of daily log returns
$\boldsymbol{\Sigma}$	Covariance matrix	Historical variability and co-movement of the assets

The simulated log returns are converted to simple returns:

$$R_{i,t} = e^{r_{i,t}} - 1$$

Symbol	Name	Description
$R_{i,t}$	Simple return	Simple return of asset i on day t
$r_{i,t}$	Log return	Daily log return of asset i on day t
e	Euler's number	Base of the natural exponential, Math input error

With equal weights $w_i = 1/5$, the portfolio's daily return is:

$$R_{p,t} = \sum_{i=1}^{5} w_i R_{i,t}$$

Symbol	Name	Description
$R_{p,t}$	Portfolio return	Equal-weight portfolio return on day t
w_i	Weight	Portfolio weight of asset i; $w_i = 1/5$
$R_{i,t}$	Simple return	Simple return of asset i on day t
$\sum$	Summation	Sum over the assets in the portfolio

The portfolio is rebalanced to equal weights each simulated day. Portfolio value is compounded through the horizon:

$$V_T = V_0 \prod_{t=1}^{T} (1 + R_{p,t})$$

Symbol	Name	Description
V_T	Final portfolio value	Value at the end of the horizon
V_0	Initial portfolio value	Starting capital, $10,000.00
$\prod$	Product	Compounds the daily growth factors $1 + R_{p,t}$
T	Horizon	Number of trading days simulated (252)

$R_{p,t}$	Portfolio return	Equal-weight portfolio return on day t

The simulation uses 10,000 paths, each with 252 trading days.

3. Main results

Metric	Result	Interpretation
Initial portfolio value	$10,000.00	Starting hypothetical capital
Mean terminal value	$13,972.82	Average terminal value across simulations
Median terminal value	$13,440.80	Middle simulated outcome
Standard deviation of terminal values	$3,930.17	Spread of terminal dollar values
1st percentile terminal value	$7,060.81	Very low modeled outcome
5th percentile terminal value	$8,547.09	Lower-tail outcome
25th percentile terminal value	$11,200.93	Lower quartile
75th percentile terminal value	$16,215.00	Upper quartile
95th percentile terminal value	$21,181.09	Upper-tail outcome
99th percentile terminal value	$25,625.84	Very high modeled outcome
Mean terminal return	39.73%	Average simulated return
Median terminal return	34.41%	Median simulated return
Probability of ending below initial value	14.45%	Fraction of paths with a loss
Probability of ending above initial value	85.55%	Fraction of paths with a gain

The percentile values describe the distribution of terminal outcomes. For example, the 5th percentile means approximately 5% of modeled outcomes are at or below that value under

the model assumptions. It is not a guarantee that future value will remain above that threshold.

4. Value at Risk and Expected Shortfall

Confidence	VaR loss estimate	Expected Shortfall loss estimate	Tail count
95%	$1,452.91	$2,323.95	500
99%	$2,939.19	$3,466.75	100

For terminal return $R_T = V_T/V_0 - 1$, the lower-tail quantile at confidence level c is:

$$q_{1-c} = Q_{1-c}(R_T)$$

Symbol	Name	Description
q_{1-c}	Lower-tail quantile	Terminal-return threshold at the lower $1-c$ tail
Q_{1-c}	Quantile function	Returns the $1-c$ quantile of a distribution
R_T	Terminal return	$V_T/V_0 - 1$, total return over the horizon
c	Confidence level	For example 95% or 99%

The script reports a signed loss amount as:

$$\mathrm{VaR}_c = -V_0\, q_{1-c}$$

Symbol	Name	Description
VaR_c	Value at Risk	Signed loss amount at confidence level c
V_0	Initial portfolio value	Starting capital, $10,000.00
q_{1-c}	Lower-tail quantile	Terminal-return threshold at the lower $1-c$ tail

Expected Shortfall is estimated from the average of returns at or below the lower-tail quantile:

$$\mathrm{ES}_c = -V_0 \, \mathbb{E}\left[R_T \mid R_T \le q_{1-c}\right]$$

Symbol	Name	Description
ES_c	Expected Shortfall	Average loss in the lower tail at confidence level c
V_0	Initial portfolio value	Starting capital, $10,000.00
$\mathbb{E}[\cdot \mid \cdot]$	Conditional expectation	Mean of terminal returns given they are at or below q_{1-c}
R_T	Terminal return	$V_T/V_0 - 1$, total return over the horizon
q_{1-c}	Lower-tail quantile	Terminal-return threshold at the lower $1-c$ tail

Here the implementation uses the empirical mean of simulated returns selected into the lower tail. In a finite sample, the exact number of observations in the tail can vary slightly due to ties and quantile conventions.

A positive reported loss amount corresponds to a loss threshold or average loss. If the modeled lower-tail quantile is positive, the signed VaR loss amount can be negative; this is mathematically possible and should not be silently changed to zero.

VaR limitation: VaR describes a loss quantile but does not indicate how severe losses can be beyond that quantile. Expected Shortfall complements it by summarizing the average loss in the selected tail.

5. What each chart tells us

1. **Terminal value distribution** — reveals the spread, central tendency, and lower/upper tails of simulated end-of-horizon values.
2. **Sample portfolio paths** — demonstrates the range of possible trajectories and how compounding can cause paths to diverge.
3. **Percentile fan chart** — compares daily percentile paths and shows how the range of modeled outcomes evolves over time.

4. **VaR versus Expected Shortfall** — compares the loss threshold with the average loss beyond that threshold at 95% and 99% confidence.
5. **Terminal return versus portfolio value** — displays individual simulated outcomes relative to the starting capital and helps identify modeled gains and losses.

The individual path chart is a sample of simulations, not all paths. The terminal value CSV contains one terminal outcome for every simulation.

6. Business and risk-management value

This analysis demonstrates how a quantitative analyst can:

- translate historical asset returns into a probabilistic scenario model;
- preserve estimated cross-asset correlation in simulated returns;
- quantify potential downside using VaR and Expected Shortfall;
- estimate the modeled frequency of ending below a capital threshold;
- communicate uncertainty through percentile charts and distributions;
- produce reproducible, auditable CSV outputs and interactive reports.

A risk manager could use such a framework as an initial scenario-analysis tool, then compare its estimates with historical stress tests, backtests, and alternative models before using it for a real risk process.

7. Model limitations and validation work still needed

- **Normality:** Multivariate normal returns can underrepresent fat tails and extreme joint market moves.
- **Stationarity:** Historical means, volatilities, and correlations may change.
- **Parameter uncertainty:** Estimated means and covariance are not known constants.
- **Regime changes:** The model does not explicitly represent crises or volatility clustering.
- **Rebalancing assumption:** Daily equal-weight rebalancing is assumed; real portfolios may rebalance less often.

- **Costs:** Transaction costs, taxes, market impact, and slippage are excluded.
- **Concentration:** Five technology-heavy stocks can remain highly exposed to common sector and market risks.
- **Model validation:** No out-of-sample calibration or VaR exceedance backtest is performed by this script.
- **Benchmark comparison:** The S&P 500 is not included as a portfolio holding in this Monte Carlo engine.

Recommended next improvements are historical or block bootstrap simulation, Student-t innovations, volatility models such as GARCH, stress scenarios, and out-of-sample backtesting.

8. Reproducibility and files

- `simulation_configuration.json` — configuration and model assumptions.
- `simulation_summary.csv` — headline statistics.
- `simulated_var_es.csv` — 95% and 99% tail risk.
- `terminal_values.csv` — all terminal simulated outcomes.
- `sample_simulation_paths.csv` — sample wealth paths.
- `sample_simulated_daily_returns.csv` — sample simulated returns.
- `charts/` — individual interactive Plotly charts.
- `monte_carlo_dashboard.html` — combined interactive dashboard.

The fixed random seed makes the random draw process reproducible for the same data, code, dependency behavior, and configuration. Minor numerical differences can occur across environments or numerical library versions.

9. Research foundation

See `docs/REFERENCES.md` for the bibliography and links to the original papers and book.

Definitions of Variables

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Monte Carlo Portfolio Risk Simulator · Educational analysis, not investment advice